

Predicting E-Commerce Customer Churn

A From-Scratch Data Preprocessing and Statistical Feature Selection Pipeline

Course: Machine Learning Practical | Problem Type: Binary Classification (Target: Churn $\in \{0,1\}$)

Source: Kaggle E-Commerce Churn Dataset (5,630 Records)

1. PROJECT TEAM & TASK OWNERSHIP MATRIX

Student Name	Roll Number	Assigned Tasks & Module Contributions
Anmol Pandey	06	B1 (Data Exploration), E1 (Label Encoding), E2 (One-Hot Encoding), F2 (Z-Score Outliers), H2 (Standardization Engine), K (Data Leakage Architecture), L (Pipeline Integration), M4 (ANOVA F-Test), M6 (Technique Matrix), M7 (Feature Selection Table), M9 (Leakage Audit), N (Audit Summary).
Drishti Nigam	56	A2 (Dataset Profiling), B2 (Descriptive Statistics Engine), D1 (ID-Scan Duplicate Detection), D2 (Inconsistent Text Cleaning), F1 (IQR Outlier Detection), H1 (Min-Max Normalization), L (Pipeline Design), M1 (Variance Threshold), M3 (Chi-Square Test of Independence), M9 (Supervised Leakage Verification), N (Comparison Table).
Abhishek Saini	02	C1 (Missing Value Analysis), C2 (Skew-Based Imputation), F3 (Outlier Treatment Logic), G (Log Transformation), I (Distribution Histograms), J (From-Scratch 80/20 Train-Test Split), L (Pipeline Sequencing), M2 (Pearson Correlation), M5 (Mutual Information Engine from Scratch), M8 (Feature Transition Audit), M9 (Feature Selection Leakage Verification), N (Final Synthesis).

2. PROBLEM DEFINITION & OPERATIONAL OBJECTIVES (TASKS A1 & A2)

In the e-commerce sector, customer churn (attrition) directly undermines recurring revenue. Acquiring new customers costs 5x to 7x more than retaining existing users. Predicting churn allows proactive customer-success outreach, tailored loyalty incentives, and logistical improvements.

TASK A1 SPECIFIC QUESTIONS & ANSWERS Problem Understanding

1. What real-world problem does the dataset represent?

The dataset represents customer attrition (churn) on an active e-commerce platform where enrolled shoppers cease future transactions.

2. Why is this problem important?

Retaining existing customers is substantially more cost-effective than customer acquisition. Predicting churn enables timely retention interventions (promotional discounts, prioritized support, customized re-engagement campaigns).

3. What is the objective of your analysis?

To construct an end-to-end, leak-free machine learning preprocessing and statistical feature-selection pipeline from mathematical first principles to accurately classify churn risk.

4. What is the target/output variable?

Churn (0 = Retained Customer, 1 = Churned Customer). Baseline churn incidence is 16.72%.

5. Is the problem Classification, Regression, Clustering, or Other?

Supervised **Binary Classification**.

TASK A2 SPECIFIC ANALYSIS Core Philosophy & Workflow Architecture

To avoid black-box machine learning, this project executes an end-to-end mathematical pipeline developed entirely from first principles:

Theory → Mathematical Formula → From-Scratch Python Implementation → Empirical Output → Statistical Interpretation

All preprocessing parameters are strictly fitted on the training split (4,504 records) and applied downstream to prevent data leakage.

3. DATASET DESCRIPTION & VARIABLE ARCHITECTURE

The dataset contains 5,630 raw observations (partitioned into 4,504 training rows and 1,126 testing rows via an 80/20 random index split). The initial schema consists of 20 columns (1 Primary Key, 1 Target Variable, and 18 Input Features: 13 Numerical, 5 Categorical).

Feature Name	Data Type	Variable Role	Description & Operational Meaning
CustomerID	Integer	Identifier	Unique account identifier for primary key tracking.
Churn	Binary Integer	Target (Output)	<i>0 = Customer Retained, 1 = Customer Churned</i> (Attrition event).
Tenure	Numerical (Float)	Input	Duration (in months) of active customer tenure on the platform.
PreferredLoginDevice	Categorical	Input	Primary login device (Phone, Mobile Phone, Computer).
CityTier	Numerical (Int)	Input	Urban classification tier of customer residence (1, 2, or 3).
WarehouseToHome	Numerical (Float)	Input	Distance (km) from closest fulfillment center to delivery location.
PreferredPaymentMode	Categorical	Input	Primary payment method (Debit Card, Credit Card, E Wallet, UPI, COD, CC).
Gender	Categorical	Input	Customer gender identity (Male, Female).
HourSpendOnApp	Numerical (Float)	Input	Average hours spent browsing the application daily.
NumberOfDeviceRegistered	Numerical (Int)	Input	Total registered devices connected to customer account (1 to 6).
PreferedOrderCat	Categorical	Input	Dominant purchasing category (Mobile, Fashion, Grocery, etc.).
SatisfactionScore	Numerical (Int)	Input	Customer satisfaction rating on a 1 (Lowest) to 5 (Highest) scale.
MaritalStatus	Categorical	Input	Marital status of customer (Single, Married, Divorced).
NumberOfAddress	Numerical (Int)	Input	Total registered shipping addresses on account.
Complain	Binary Integer	Input	Customer raised a complaint within the past month (<i>1 = Yes, 0 = No</i>).

Feature Name	Data Type	Variable Role	Description & Operational Meaning
OrderAmountHikeFromlastYear	Numerical (Float)	Input	Percentage growth in spending compared to prior year.
CouponUsed	Numerical (Float)	Input	Total promotional discount coupons redeemed in the past year.
OrderCount	Numerical (Float)	Input	Total count of completed purchase transactions.
DaySinceLastOrder	Numerical (Float)	Input	Recency in days since customer's most recent order.
CashbackAmount	Numerical (Float)	Input	Average cashback reward (\$) credited per order.

4. FROM-SCRATCH DESCRIPTIVE STATISTICAL ENGINE (TASK B2)

To profile dispersion and central tendencies without relying on automated library functions, statistical parameters were calculated manually across all 13 numerical inputs:

$$\mu = (1/n)\Sigma X_i \mid \sigma^2 = (1/n)\Sigma (X_i - \mu)^2 \mid \sigma = \sqrt{\sigma^2} \mid Range = X_{max} - X_{min}$$

Feature	Mean	Median	Mode	Min	Max	Range	Variance (σ²)	Std Dev (σ)
Tenure	10.25	9.00	1.00	0.00	61.00	61.00	73.59	8.58
WarehouseToHome	15.68	14.00	9.00	5.00	127.00	122.00	73.48	8.57
HourSpendOnApp	2.92	3.00	3.00	0.00	5.00	5.00	0.52	0.72
NumberOfDeviceRegistered	3.68	4.00	4.00	1.00	6.00	5.00	1.03	1.02
NumberOfAddress	4.20	3.00	2.00	1.00	21.00	20.00	6.63	2.58
OrderCount	3.01	2.00	2.00	1.00	16.00	15.00	8.74	2.96
CouponUsed	1.74	1.00	1.00	0.00	15.00	15.00	3.49	1.87
DaySinceLastOrder	4.54	3.00	3.00	0.00	46.00	46.00	13.50	3.67
CityTier	1.66	1.00	1.00	1.00	3.00	2.00	0.84	0.92
SatisfactionScore	3.07	3.00	3.00	1.00	5.00	4.00	1.92	1.39
OrderAmountHikeFromlastYear	15.70	15.00	13.00	11.00	26.00	15.00	13.52	3.68
Complain	0.28	0.00	0.00	0.00	1.00	1.00	0.20	0.45
CashbackAmount	177.04	163.23	149.36	0.00	324.99	324.99	2424.58	49.24

5. DATA QUALITY AUDITING, MISSINGNESS & CLEANING

5.1 Missing Value Diagnostics (Task C1) & Justified Imputation (Task C2)

Missing value scanning identified 1,478 total null entries across exactly 7 numerical columns (~4.40% to 5.26% missingness per feature). Custom imputation rules were applied based on distributional skewness and feature semantics:

TASKS C1 & C2 SPECIFIC QUESTIONS & ANSWERS Imputation Strategy & Justification

Q: Why was a specific method chosen for each missing feature?

- **Median Imputation (Skewed Continuous Data):** Applied to `Tenure`, `WarehouseToHome`, `OrderAmountHikeFromlastYear`, and `DaySinceLastOrder` because extreme values (e.g., `WarehouseToHome` max = 127 km) pull the mean away from the true center. The median reflects the 50th percentile rank and is robust to outliers.
- **Mean Imputation (Symmetric Distribution):** Applied to `HourSpendOnApp`, which displays an approximately normal bell distribution with mean (2.92) $\approx$ median (3.00).
- **Mode Imputation (Discrete Counts):** Applied to count variables `CouponUsed` and `OrderCount` to ensure imputed values reflect valid discrete transaction integers.

Feature Name	Missing Count	Missing %	Distributional Shape	Selected Metric	Pre-Impute Nulls	Post-Impute Nulls
DaySinceLastOrder	237	5.26%	Right-Skewed Tail (Max = 46.0)	Median (3.0)	237	0
Tenure	218	4.84%	Heavy Right Skew (Max = 61.0)	Median (9.0)	218	0
OrderAmountHikeFromlastYear	212	4.71%	Moderately Right-Skewed	Median (15.0)	212	0
WarehouseToHome	206	4.57%	Heavy Outlier Tail (Max = 127.0)	Median (14.0)	206	0
CouponUsed	205	4.55%	Discrete Count Feature	Mode (1.0)	205	0
OrderCount	202	4.48%	Discrete Count Feature	Mode (2.0)	202	0
HourSpendOnApp	198	4.40%	Symmetric (Mean = 2.92, Med = 3.0)	Mean (2.92)	198	0

5.2 Duplicate Verification (Task D1) & Categorical Cleanup (Task D2)

TASKS D1 & D2 SPECIFIC QUESTIONS & ANSWERS Duplicate Risks & Data Consistency

Q: Why do duplicate observations create problems, should all duplicates be deleted, and what decision was made?

- **Problems caused by duplicates:** Duplicates artificially inflate statistical significance, bias models toward repeated observations, and cause data leakage if copies exist in both training and test sets.
- **Deletion rationale:** Exact primary key replicates must be eliminated, whereas non-identical records sharing natural feature combinations must be retained.
- **Decision & Finding:** A from-scratch hash-set scan on `CustomerID` confirmed **0 duplicate records** across the 4,504 training rows (100% record integrity).

Q: How were inconsistent category spellings, capitalization, and whitespace handled?

- **PreferredLoginDevice:** Standardized `Mobile Phone` $\rightarrow$ `Phone` (collapsed from 3 $\rightarrow$ 2 unique categories: `Phone`, `Computer`).
- **PreferredPaymentMode:** Standardized `CC` $\rightarrow$ `Credit Card`, `COD` $\rightarrow$ `Cash On Delivery`, `E wallet` $\rightarrow$ `E Wallet`, `UPI` $\rightarrow$ `Upi` (collapsed from 7 $\rightarrow$ 5 unique categories).
- **PreferredOrderCat:** Standardized `Mobile Phone` $\rightarrow$ `Mobile` (collapsed from 6 $\rightarrow$ 5 categories).
- **Boundary Validation:** Checked all 13 numerical inputs; confirmed 0 negative or impossible values.

6. CATEGORICAL ENCODING, OUTLIER ANALYSIS & SCALING

6.1 From-Scratch Encoding Implementations (Tasks E1 & E2)

TASKS E1 & E2 SPECIFIC QUESTIONS & ANSWERS Encoding Theory & Pitfalls

Q: What is Label Encoding, when is it suitable, and what problem arises if applied to nominal categories?

Label Encoding maps distinct categories directly to integers. It is suitable strictly for **binary features** or **ordinal variables** where relative rank is meaningful (e.g. Low=0, Med=1, High=2). When applied to nominal multiclass categories (e.g. Payment Mode: Credit Card=0, UPI=1, COD=2), algorithms misinterpret the assigned integers as continuous quantities ($COD > UPI > Credit Card$), imposing false geometric distance.

Q: Why is One-Hot Encoding useful, what is the difference, and when should each be used?

One-Hot Encoding (OHE) creates an independent binary dummy indicator (0 or 1) for each category level, preventing false ordinal ranking. In this pipeline, Label Encoding was applied solely to binary Gender ({'Female': 0, 'Male': 1}), while OHE was implemented from scratch for 4 multiclass nominal columns (PreferredLoginDevice , PreferredPaymentMode , PreferredOrderCat , MaritalStatus), generating 15 binary dummy indicator features and expanding feature dimensionality from 20 to 36.

6.2 Outlier Analysis: IQR vs. Z-Score Comparison (Tasks F1 & F2)

Outliers were detected from scratch using Interquartile Range ($Q_1 - 1.5 \times IQR$, $Q_3 + 1.5 \times IQR$) and Z-Score ($|Z| > 3.0$).

Feature Name	Q1	Q3	IQR	IQR Bounds [Lower, Upper]	IQR Outliers Detected	Z-Score Outliers (Z > 3.0)
OrderCount	1.00	3.00	2.00	[-2.00, 6.00]	567	148
CouponUsed	1.00	2.00	1.00	[-0.50, 3.50]	504	88
CashbackAmount	145.80	196.19	50.39	[70.22, 271.77]	351	10
NumberOfDeviceRegistered	3.00	4.00	1.00	[1.50, 5.50]	310	0
DaySinceLastOrder	2.00	7.00	5.00	[-5.50, 14.50]	52	37
OrderAmountHikeFromlastYear	13.00	18.00	5.00	[5.50, 25.50]	27	0
HourSpendOnApp	2.00	3.00	1.00	[0.50, 4.50]	6	3
Tenure	3.00	15.00	12.00	[-15.00, 33.00]	3	3
WarehouseToHome	9.00	20.00	11.00	[-7.50, 36.50]	2	2
NumberOfAddress	2.00	6.00	4.00	[-4.00, 12.00]	2	2
CityTier / SatisfactionScore / Complain	Bounded Discrete Inputs				0	0
Total Outlier Instances	Cumulative Count				1,824	293

TASK F2 & F3 SPECIFIC QUESTIONS & ANSWERS Why Counts Differ & Outlier Treatment Logic

Q: Why do IQR and Z-score methods produce different outlier counts?

IQR (1,824 outliers) uses percentile ranks, isolating extreme tails in narrow discrete columns without parametric assumptions. Z-Score (293 outliers) relies on sample standard deviation σ ; extreme outliers inflate σ , widening the $\pm 3\sigma$ threshold and creating a **masking effect** on moderate anomalies.

Q: Task F3 Outlier Decision — Removed, Capped, Replaced, Transformed, or Retained?

Decision: Retained & Transformed via Log Transformation ($\ln(X+1)$).

Justification: Dropping 1,824 rows would purge ~40.5% of the dataset, eliminating valid high-value customers. Instead of deletion or artificial capping, a natural log transformation ($X' = \ln(X + 1)$) was applied from scratch to `CashbackAmount` and `OrderAmountHikeFromLastYear`. This compressed positive skewness while preserving exact relative ordering and variance structure.

6.3 Feature Scaling: Normalization vs. Standardization (Tasks H1 & H2)

TASKS H1 & H2 SPECIFIC QUESTIONS & ANSWERS Scaling Comparison & Use Cases

Q: What is the difference between Normalization and Standardization, and when should each be used?

- **Min-Max Normalization** ($X' = (X - X_{min}) / (X_{max} - X_{min})$): Bounds features strictly into $[0, 1]$. Best suited for bounded algorithms (e.g. Neural Networks, KNN, Computer Vision pixel inputs). Sensitive to outliers.
- **Standardization** ($Z = (X - \mu) / \sigma$): Centers features at $\mu = 0.0$, $\sigma = 1.0$. It does not bind data into fixed ranges and handles outliers effectively. Best for Logistic Regression, SVMs, and gradient descent optimization.

Both scaling engines were implemented from scratch and matched scikit-learn outputs with 100% precision.

7. DATA LEAKAGE PREVENTION & STATISTICAL FEATURE SELECTION

7.1 Leakage Architecture & Train-Test Partitioning (Tasks J, K & M9)

TASKS J, K & M9 SPECIFIC QUESTIONS & ANSWERS Train-Test Split & Leakage Prevention

1. **Why do we divide data into training and testing sets?** To evaluate generalization performance on unseen real-world records.
2. **Why should the same observations not be used for training and evaluation?** Testing on training data causes overfitting and inflated validation metrics that fail in production.
3. **What is the role of randomization?** Random index shuffling ensures train and test sets follow identical probability distributions.
4. **Why fitting preprocessing on the complete dataset causes Data Leakage:** Learning global parameters (μ_{all} , σ_{all} , or supervised selection) leaks test information into training. **Pipeline Isolation Rules:** (1) Split executed first (80/20); (2) Imputation parameters learned strictly from train; (3) Scaling bounds learned from train; (4) Supervised statistical tests evaluated solely on training target labels.

7.2 Method Selection Matrix (Task M6)

Method	Input Type	Target	Mathematical Purpose
Variance Threshold	Numerical / OHE	None	Detect constant or quasi-constant uninformative features.
Pearson Correlation (r)	Numerical	Target / Numerical	Measure linear relationship and inter-feature redundancy.
Chi-Square (χ^2)	Categorical	Categorical	Test categorical independence against churn.
ANOVA F-Test	Numerical	Categorical	

Method	Input Type	Target	Mathematical Purpose
			Compare between-group vs. within-group variance across churn.
Mutual Information	Discrete / Categorical	Categorical	Quantify total shared non-linear information in bits.

TASKS M1, M2 & M4 SPECIFIC QUESTIONS & ANSWERS Statistical Concepts & Non-Linearity

- **Task M1 (Variance Threshold):** Near-zero variance features provide limited discriminatory information because they remain constant across nearly all churn and retention instances.
- **Task M2 (Pearson Correlation):** Measures strictly linear correlation. *Important:* Low Pearson r does not imply no relationship (e.g. WarehouseToHome has $r = +0.070$, but strong non-linear ANOVA $F = 22.15$).
- **Task M4 (ANOVA F-Test):** High F-statistic indicates group means differ significantly relative to internal variation. Assumptions: Normality, homogeneity of variance, and observation independence.

7.3 Mutual Information Engine: From-Scratch Implementation & Verification (Task M5)

Mutual Information (MI) measures the mutual dependence between a discrete/categorical feature X and target $Y = \text{Churn}$ in information bits using base-2 logarithms: $H(Y) = -\sum_y P(y) \log_2 P(y)$, $H(Y|X) = \sum_x P(x) H(Y|X=x)$, $MI(X; Y) = H(Y) - H(Y|X)$.

TASK M5 SPECIFIC QUESTIONS & ANSWERS Theoretical Foundations of Mutual Information

- 1. Why is Mutual Information measured in bits when $\log_2$ is used?**
Using the base-2 logarithm scales entropy to binary decision units (bits). One bit represents the amount of information required to resolve the uncertainty between two equally probable outcomes.
- 2. What does $MI(X; Y) = 0$ mean?**
It proves that X and Y are statistically independent under the estimated joint distribution. Knowing the feature provides zero reduction in uncertainty about customer churn.
- 3. What does $MI(X; Y) = H(Y)$ mean?**
It means that $H(Y|X) = 0$. Knowing feature X completely eliminates all entropy/uncertainty about customer churn, making classification deterministic.
- 4. How is MI different from Pearson correlation and can it identify non-linear relationships?**
Pearson correlation measures strictly linear associations ($r \in [-1, +1]$). Mutual Information quantifies general probabilistic dependence, detecting complex non-linear, non-monotonic, and multi-modal relationships where Pearson correlation evaluates to zero.
- 5. Why is MI estimation difficult for continuous variables?**
For continuous variables, naive small frequency binning introduces substantial discretization bias. Exact continuous MI calculation requires differential entropy integration ($\iint p(x,y) \log[p(x,y)/(p(x)p(y))] dx dy$) requiring specialized non-parametric nearest-neighbor estimators (such as the Kraskov-Stögbauer-Grassberger / KSG algorithm) rather than simple contingency counting.

Worked Example from Scratch: Complain vs. Churn

- **Overall Target Entropy:** $H(\text{Churn}) = -[P(0) \log_2 P(0) + P(1) \log_2 P(1)] = 0.6512 \text{ bits}$ (where $P(0) = 0.8328$, $P(1) = 0.1672$).
- **Conditional Entropy:** For Complain=0 ($P(X)=0.7200$, $H(Y|X=0)=0.4992$, contributes 0.3595); for Complain=1 ($P(X)=0.2800$, $H(Y|X=1)=0.8987$, contributes 0.2516). Total $H(\text{Churn} | \text{Complain}) = 0.6111 \text{ bits}$.
- **From-Scratch MI:** $MI = H(Y) - H(Y|X) = 0.6512 - 0.6111 = 0.0402 \text{ bits}$. Matches scikit-learn's `mutual_info_score` converted to bits with 100% precision.

Feature Name	H(Y) — Target Entropy (bits)	H(Y X) — Conditional Entropy (bits)	Mutual Information (bits)
Complain	0.6512	0.6111	0.0402
PreferredOrderCat	0.6512	0.6118	0.0394
MaritalStatus	0.6512	0.6276	0.0236
SatisfactionScore	0.6512	0.6412	0.0100
PreferredPaymentMode	0.6512	0.6449	0.0063
CityTier	0.6512	0.6469	0.0044
PreferredLoginDevice	0.6512	0.6487	0.0025
Gender	0.6512	0.6505	0.0007

7.4 Multi-Criteria Feature Selection Decision Table (Tasks M1–M7)

$$r = \Sigma((x - \bar{x})(y - \bar{y})) / \sqrt{[\Sigma(x - \bar{x})^2 \Sigma(y - \bar{y})^2]} \quad | \quad \chi^2 = \Sigma(O - E)^2 / E \quad | \quad F = [SSB / (k - 1)] / [SSW / (N - k)]$$

Feature Name	Data Type	Variance	Pearson (r)	ANOVA (F)	Chi-Sq (χ²)	Decision	Empirical Justification
Tenure	Numerical	73.59	-0.342	591.20	—	KEEP	Strongest predictor overall; high tenure lowers churn risk.
Complain	Binary Num	0.20	+0.250	291.98	274.19	KEEP	Complaining users churn at 31.5% vs. 11.0% non-complainants.
DaySinceLastOrder	Numerical	13.50	-0.160	116.83	—	KEEP	High recency latency directly drives customer churn.
CashbackAmount	Numerical	2424.58	-0.154	108.45	—	KEEP	Higher cashback provides financial retention incentive.
SatisfactionScore	Numerical	1.92	+0.105	49.97	—	KEEP	Significant variance across satisfaction ratings.
NumberOfDeviceRegistered	Numerical	1.03	+0.108	52.79	—	KEEP	High devices correlate with shared/churned accounts.

Feature Name	Data Type	Variance	Pearson (r)	ANOVA (F)	Chi-Sq (χ^2)	Decision	Empirical Justification
CityTier	Numerical	0.84	+0.085	32.70	27.92	KEEP	Tier 3 urban centers exhibit higher churn friction.
WarehouseToHome	Numerical	73.48	+0.070	22.15	—	KEEP	Distance impacts delivery latency; ANOVA $F > 3.84$.
NumberOfAddress	Numerical	6.63	+0.044	8.78	—	KEEP	Significant variation across churn groups ($F = 8.78$).
OrderCount	Numerical	8.74	-0.029	3.89	—	KEEP	Exceeds critical F threshold (3.84); baseline order volume.
MaritalStatus	Categorical	—	—	—	154.81	KEEP	High statistical dependency; singles show elevated churn.
PreferredOrderCat	Categorical	—	—	—	242.62	KEEP	Product category dictates repeat purchasing cycles.
PreferredPaymentMode	Categorical	—	—	—	41.85	KEEP	Payment friction (COD vs. Credit Card) impacts retention.
PreferredLoginDevice	Categorical	—	—	—	16.16	KEEP	Statistically dependent on churn ($\chi^2 = 16.16, p < 0.001$).
Gender	Categorical	0.24	+0.029	3.85	4.24	KEEP	Statistically significant ($\chi^2 = 4.24 > 3.841, p < 0.05$).
HourSpendOnApp	Numerical	0.52	+0.019	1.66	—	REMOVE	ANOVA $F (1.66) < 3.84$; Pearson $r \approx 0$. No signal.
CouponUsed	Numerical	3.49	-0.008	0.32	—	REMOVE	ANOVA $F (0.32)$ fails significance; near-zero correlation.

Feature Name	Data Type	Variance	Pearson (r)	ANOVA (F)	Chi-Sq (χ²)	Decision	Empirical Justification
OrderAmountHikeFromlastYear	Numerical	13.52	-0.0002	0.0001	—	REMOVE	$F \approx 0$ and $r \approx 0$; statistically independent of churn.

8. FINAL PIPELINE RESULTS & COMPARATIVE AUDIT (TASK N)

8.1 Feature Dimensionality Flow (Task M8)

Pipeline Execution Stage	Count	Stage Definition
1. Original Dataset	19 Features	18 input features + 1 target (Churn)
2. After Preprocessing	35 Features	13 numerical + 1 label encoded + 21 OHE dummy features
3. After Feature Selection	26 Features	Statistically validated features (3 uninformative inputs removed)

8.2 Complete Preprocessing Comparison Table (Task N)

Parameter	Before Preprocessing	After Preprocessing
Records	5,630	4,504
Features	19	35
Missing Values	1,478	0
Duplicate Records	0	0
Categorical Features	5	0
Outliers (IQR Method)	1,824	1,824
Outliers (Z-Score Method)	293	293
Selected Features	18	26

TASK N SPECIFIC QUESTIONS & ANSWERS Detailed Interpretation of Audit Parameters

- **Records (5,630 → 4,504):** Preprocessing and modeling transformations are executed strictly on the 80% training split (4,504 records) to prevent test-set data leakage.
- **Features (19 → 35):** Raw columns expanded as 5 categorical variables were transformed into 1 label-encoded column and 21 One-Hot Encoded binary indicators.
- **Missing Values (1,478 → 0):** Fully resolved across all 7 numerical features using skewness-aligned Median, Mean, and Mode imputation without deleting rows.
- **Categorical Features (5 → 0):** 100% of raw text categorical features were converted into machine-readable numerical representations.
- **Outliers (1,824 IQR / 293 Z-Score):** 100% retained and treated via non-destructive $\ln(X+1)$ transformation, avoiding sample truncation while reducing positive skewness.
- **Selected Features (18 → 26):** After expanding to 35 features, statistical elimination of 3 non-significant inputs (HourSpendOnApp , CouponUsed , OrderAmountHikeFromlastYear) finalized the pipeline at 26 validated model-ready features.

9. KEY ANALYTICAL FINDINGS & PRACTICAL INSIGHTS

- 1. Customer Complaints are the Dominant Churn Driver:** Customers filing complaints exhibit a 31.5% churn rate versus 11.0% for non-complainants ($\chi^2 = 274.19$, $ANOVA F = 291.98$). Automated escalation workflows for customer complaints represent the single highest-ROI retention initiative.
- 2. Tenure Protects Against Churn:** Customer tenure displayed the strongest overall statistical relationship ($ANOVA F = 591.20$, $Pearson r = -0.342$). Newly onboarded users are most vulnerable; dedicated onboarding campaigns in months 1–3 are crucial.
- 3. Linear Correlation Alone is Insufficient:** Low linear correlation features such as WarehouseToHome ($r = +0.070$) and CashbackAmount ($r = -0.154$) demonstrated strong non-linear significance under ANOVA ($F = 22.15$ and 108.45). Feature selection must incorporate multi-test statistical validation.
- 4. Justified Elimination of Irrelevant Features:** HourSpendOnApp ($F = 1.66$), CouponUsed ($F = 0.32$), and OrderAmountHikeFromLastYear ($F = 0.0001$) were proven statistically independent of churn and safely eliminated, reducing model complexity and training latency.

Conclusion: By combining from-scratch mathematical preprocessing with leak-free statistical feature selection, the raw e-commerce dataset was transformed into a robust, clean, and statistically validated 26-feature dataset primed for high-performance machine learning classification.